

Synthetic Market Phase 20

Paired denoise/noise robustness battery for the Leader (L4) and Dispersion (L35) market subspaces. This report summarizes the 20-cell-per-hypothesis matrix, emphasizing matched random controls, lambda sweeps, and what the resulting evidence does and does not support.

DATE	SAMPLE	GENERATION	ENGINE
28 March 2026	1280 requests across 2 hypotheses	batch=32, max_tokens=15000, paired denoise/noise	compiled for baseline/project-out, eager fallback for random control

MAIN READ

Phase 20 is materially stronger than the earlier exploratory batches: it uses matched paired cohorts, denoise and noise directions, a lambda sweep, and matched orthogonal random controls. The main empirical result is clean and consistent: project_out is *less disruptive than matched random control in all 12 / 12* tool-token and generated-token selectivity comparisons. Mechanically the patching is also clean (1152 / 1152 applied, zero skips). The main remaining methodological gap is that source behaviors were not generated, so this is still *paired robustness evidence*, not a full source-restoration mediation demonstration.

SELECTIVITY WINS	TOKEN-DELTA WINS	PATCH COVERAGE	METHODOLOGY
12 / 12	12 / 12	1152 / 1152	Paired+
tool-token gap comparisons won by project_out versus random control	generated-token delta comparisons won by project_out	all patch entries applied; no skipped rows	strong robustness battery, still short of full restoration mediation

Methodology Scorecard

Best-practice item	Status
Matched paired prompt cohorts (denoise + noise)	meets

Best-practice item	Status
Lambda sweep over intervention strength	meets
Matched orthogonal random-direction controls	meets
Exact span targeting with per-row patch diagnostics	meets
Source-behavior restoration metrics from generated source outputs	missing
Neighboring-component / neighboring-PC sweep	missing
Subspace-size sweep	missing
Path-level validation	missing
Norm/distribution diagnostics	partial

Phase 20 is best described as:

- A paired robustness battery over Leader (L4) and Dispersion (L35) subspaces
- With denoise and noise cohorts, lambda sweep {0.5, 1.0, 1.5}, and matched orthogonal random controls
- Backed by explicit per-row patch coverage and norm diagnostics

It is a substantially better causal test than the earlier single-condition ablations, but it still lacks source-behavior restoration metrics and path validation.

Experimental Design

Matrix Layout

Parameter	Value
Hypotheses	Leader (vol_1h_max, L4) and Dispersion (pct_1h_mad, L35)
Rows per cell	32
Cells per hypothesis	20: 2 baselines + 6 project-out + 12 random control
Pair modes	denoise, noise
Lambda sweep	0.5, 1, 1.5
Random-control seeds	11, 17
Batch size / max tokens	32 / 15000
Runtime note	baseline + project_out on compiled custom-op path; random_control on eager fallback because compiled support is currently project_out-only

What denoise And noise Mean Here

A pair mode label just tells you *which member of a matched prompt pair we intervene on*.

For each hypothesis, the code builds pairs of similar prompts from the same family / variant / roster group, then sorts the two prompts by the visible metric:

- Leader: vol_1h_max

- Dispersion: pct_1h_mad

Then:

- denoise means: *start from the lower-metric prompt in the pair*
- noise means: *start from the higher-metric prompt in the pair*

So in plain English:

- denoise asks, “what happens if we intervene on the weaker / lower-signal prompt?”
- noise asks, “what happens if we intervene on the stronger / higher-signal prompt?”

What these labels do *not* mean in Phase 20:

- they do not mean we literally copy activations from one prompt into the other
- they do not mean we already have a full clean→corrupt or corrupt→clean restoration test

In this phase, the paired prompt only tells us which side of the matched pair we are testing. The actual intervention on that prompt is still one of:

- no patch baseline
- targeted project_out
- matched orthogonal random_control

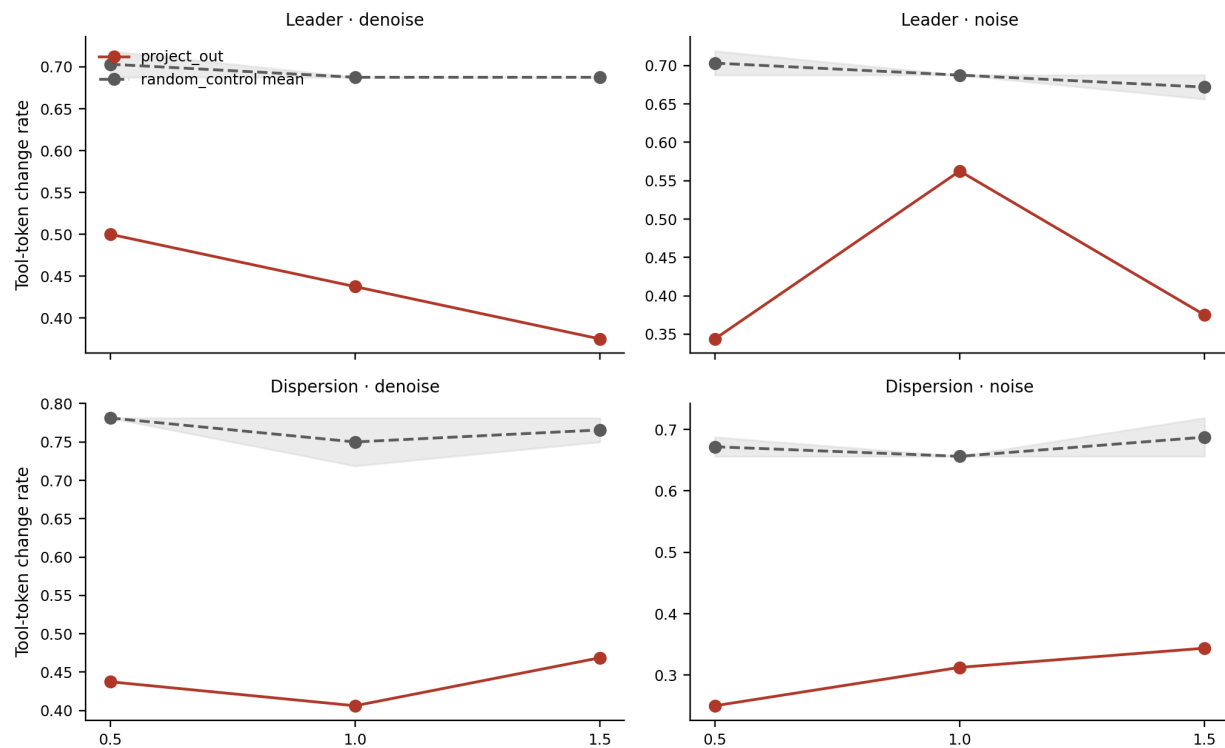
Patch Sanity

Hypothesis	Entries	Applied	Coverage	Std ratio	Norm ratio
Leader	576	1	1	0.999	1.002
Dispersion	576	1	1	0.999	0.999

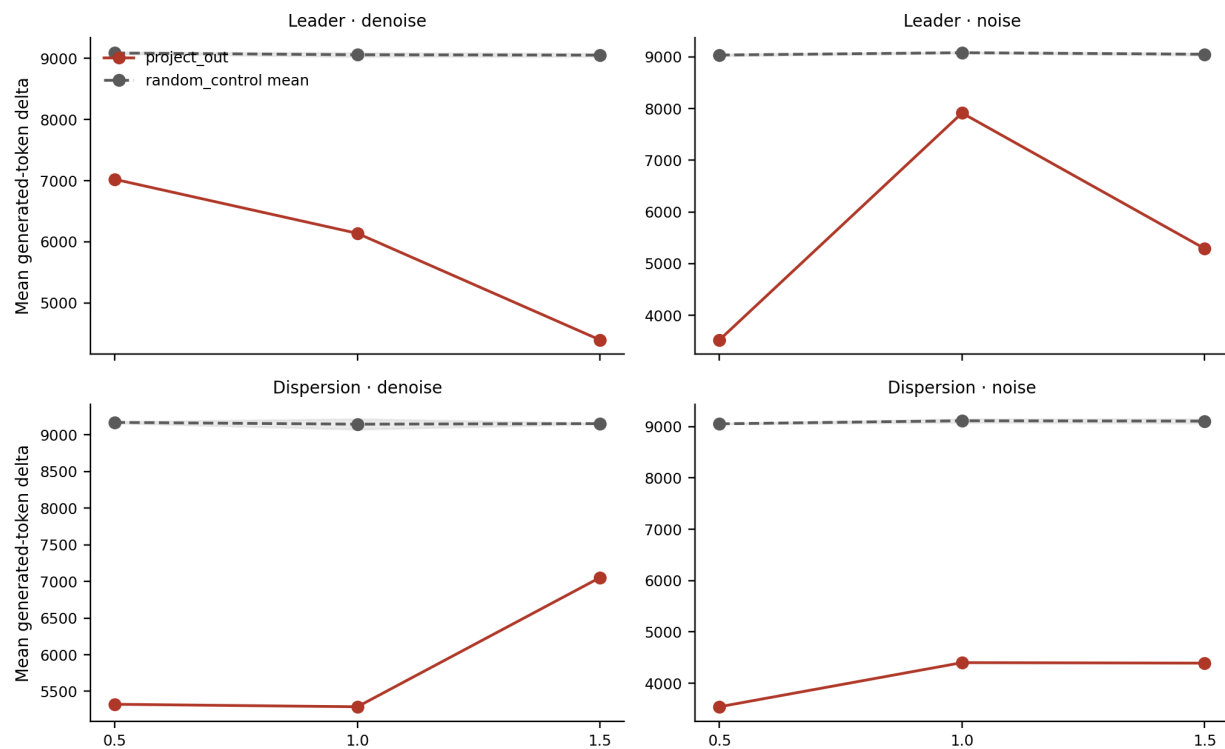
The intervention is mechanically clean:

- patch_applied_rate = 1.0 for both hypotheses
- patch_skipped_rate = 0.0 for both hypotheses
- coverage is full
- average norm ratios stay close to 1.0

Behavioral Results

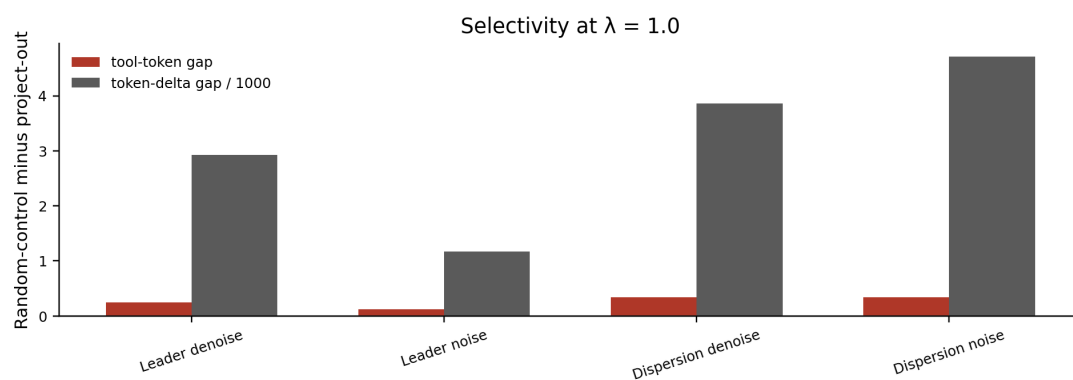


Tool-token change rates across the lambda sweep. In every panel, `project_out` stays below the matched random-control mean.

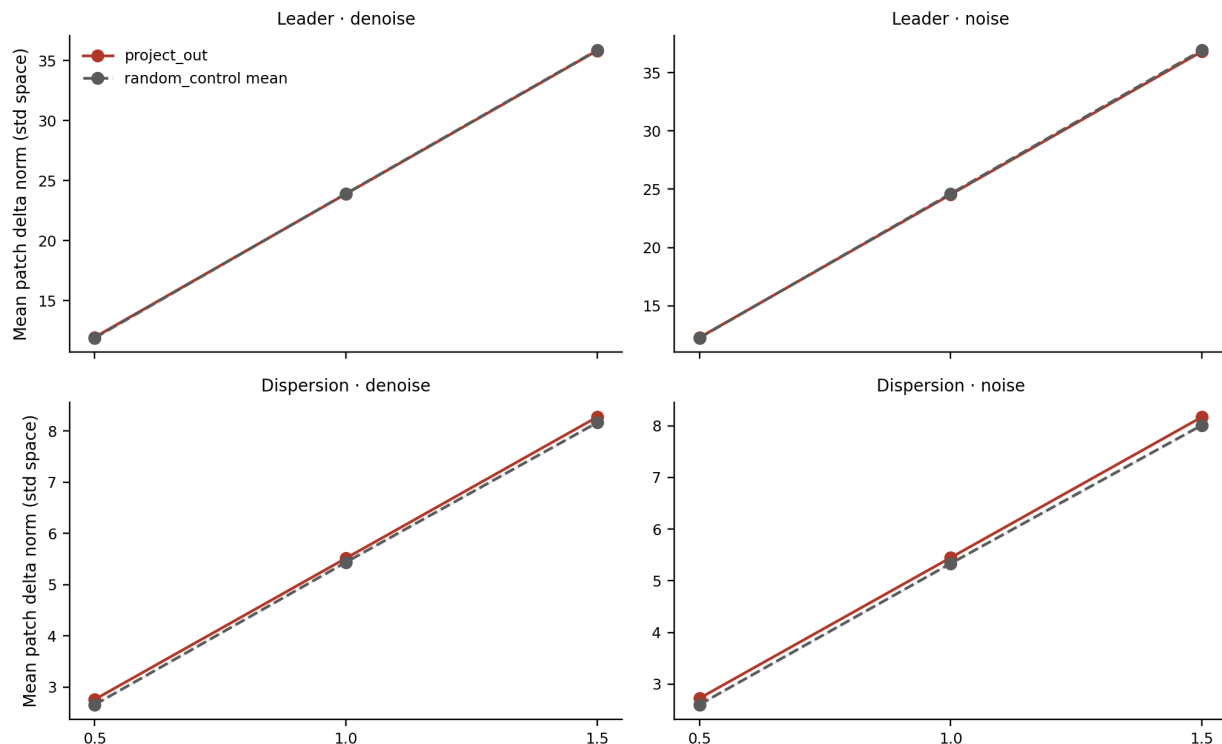


Generated-token deltas show the same pattern: targeted project-out remains less disruptive than matched random control in all 12 comparisons.

Cell	Tool-token gap	Tool-name gap	Token-delta gap
Leader denoise, $\lambda = 1.0$	0.25	0.219	2921.3
Leader noise, $\lambda = 1.0$	0.125	0.094	1166.9
Dispersion denoise, $\lambda = 1.0$	0.344	0.281	3856.3
Dispersion noise, $\lambda = 1.0$	0.344	0.344	4715.8



Positive values mean random control is more disruptive than targeted project-out. The effect is strongest for the dispersion hypothesis.



Patch delta norms scale with λ as expected, while the mean hidden-state norm stays close to the unpatched baseline on average.

The empirical story is consistent:

- project_out wins every tool-token and generated-token selectivity comparison (12 / 12 each)
- the largest $\lambda = 1.0$ tool-token gaps appear in the dispersion cells (0.344 in both denoise and noise)
- the generated-token selectivity gap is also strongest for dispersion, especially noise (4715.8 tokens)
- the effect is selective rather than explosive: patch norms scale with λ , but overall hidden-state norms remain near baseline

How To Read The Random-Control Comparison

The key comparison in Phase 20 is not “does project_out cause more damage than random control?” It is the opposite question:

- does the targeted intervention behave differently from a matched arbitrary perturbation?

Here, “matched” means the control uses:

- the same prompt cohort
- the same layer and token span
- the same number of removed dimensions
- a comparable intervention scale

What we see is that project_out is *consistently less disruptive* than the matched random control. For example, at $\lambda = 1.0$:

- Leader denoise: tool-token change gap = 0.250

- Leader noise: tool-token change gap = 0.125
- Dispersion denoise: tool-token change gap = 0.344
- Dispersion noise: tool-token change gap = 0.344

This matters because it rules out a weak interpretation:

- “maybe any hidden-state perturbation of this size would produce the same behavioral effect”

If that weak interpretation were true, the random controls should look similar to the targeted patch. Instead, the random controls are *more destructive* on every paired comparison. The most natural reading is:

- the discovered subspace is behaviorally special
- the targeted intervention is more selective and more on-manifold than random damage
- the observed effect is not just generic corruption

Just as important, this result should *not* be overread. “Less disruptive than random control” does *not* mean:

- the targeted subspace is the unique causal mediator
- the intervention is stronger than random control
- the full causal mechanism has been identified

The right interpretation is narrower:

- the subspace appears non-arbitrary and behaviorally meaningful
- removing it changes behavior in a structured way
- random perturbations produce larger off-manifold damage than the targeted project-out

Interpretation

Phase 20 supports a stronger claim than Phase 19:

1. The targeted market subspaces are not interchangeable with matched random directions. Across both hypotheses, both pair modes, and all tested lambda values, the targeted intervention is systematically less disruptive than orthogonal controls.
2. That makes the subspaces look behaviorally special and more on-manifold than random perturbations.
3. The paired denoise/noise structure also reduces the risk that the earlier result was just a one-direction artifact.

But the report should still stay disciplined:

- source behaviors were not generated, so there is no direct restoration score
- neighboring-component and subspace-size sweeps were not run in this matrix
- there is still no path-level validation showing that the behavior flows through the hypothesized route
- because the targeted intervention is *less* disruptive than random control, this is best read as evidence of selectivity and non-arbitrariness, not as evidence that we have fully ablated the model’s only market representation

The right label is therefore: *strong paired robustness evidence for behaviorally meaningful market subspaces*, not yet *a definitive mediation result for a unique causal variable*.

Next Steps

To close the remaining methodological gaps:

1. Generate explicit source behaviors for the paired cells so the analysis can report restoration and backfire rates directly.
2. Add neighboring-component and subspace-size sweeps around the selected top-4 basis.
3. Extend the compiled custom-op path beyond `project_out` so all matrix cells can share the same runtime profile.
4. Add path-level validation or finer-grained site sweeps to test whether the effect localizes to the intended computational route.